Babäk Firoozi Fooladi

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Data analyst

Doctoral candidate and data analytics engineer with six years of experience in academic and consultancy settings, where they built insight-driven dashboards, strong data models, and end-to-end data pipelines with carefully curated visualisations. I have extensive expertise developing ETL/ELT workflows and converting complicated datasets into understandable, practical insights for stakeholders. I have extensive experience working with teams from many departments to establish key performance indicators, provide data for decision-making, and implement analytics solutions that have a direct impact on strategy.

Technical Skills

Python:	●●●	**PostgreSQL:**	●●●	**DuckDB:**	●●●●	**SQL:**	●●●
Spatial data:	●●●●●	**LLM/NLP/LLM:**	●●●●	**SQLite:**	●●●	**AZURE:**	●●
GCP:	●●●●	**R:**	●●●●	**MongoDB:**	●●	**Excel:**	●●●●
AcrGIS:	●●●	**ETL/ELT:**	●●●●	**Cartography:**	●●●●	**STATA:**	●●●●
Git:	●●●	**RAG:**	●●●●●	**Julia:**	●●	**PowerBI:**	●●

Soft Skills

Team work:	●●●	**Self-organised:**	●●●	**Project management:**	●●●●
Accounting:	●●	**Quality control:**	●●●●	**Visual communication:**	●●●●●
Quantiative research:	●●●●	**Prompt engineering:**	●●●●	**Document Management:**	●●●●

Experience

DATA ANALYTICS & ENGINEERING — Jan 2025 – Present
Self-Employed | Consultant — *Project-Based – Espoo, Finland*
Delivering end-to-end data engineering, analytics, and BI solutions tailored to each client's industry and business objectives. Advising on data strategy with a focus on scalability and sustainable growth. As a data specialist, I design efficient pipelines that handle diverse data types and translate complex data into clear, actionable insights aligned with client goals.

Building Information Backend — Jan 2026 – Present
Apex-Heat – Finland

- Consulting on data transformation and business logic to facilitate future integration of spatial data with third-party systems.
- Developing a data pipeline to transform and compute key metrics from building spatial data across Finland.
- Designing and implementing an **API-based pipeline** for automated building calculations in Finland.
- Evaluating and structuring spatial data models for EU-wide coverage.
- Modelling data leveraging domain expertise in real estate economics and zoning regulations.

Optimisation & Technical Debt Reduction — Jan 2025 – Dec 2025
Apex-Heat – Finland

- Identified and resolved **technical debt**, significantly improving data pipeline performance and maintainability.
- Led migration from **Pandas** to **Polars** (including GeoPandas), reducing computation time and improving overall pipeline efficiency.
- Refactored codebase design patterns and produced comprehensive **technical documentation**.
- Expanded and improved **test coverage** and testing methodologies for data processing workflows.

Ph.D. Candidate Sep 2020 – Present
Aalto University *Full-time – Espoo, Finland*
Conducting doctoral research in urban and real estate economics, applying econometric and machine learning techniques to housing markets, real estate development, and urban planning. Managing the full spatial data lifecycle across research projects, from collection and processing to analysis and visualization, while building and maintaining the team's data infrastructure.

- Preparing and maintaining **data visualizations** for academic presentations and lectures.
- Designing data-pipelines for local AI to extract features from text and scrctured data.
- Providing IT service and assitance to my research team to resolve issues with their computers and software.
- Applying Econometric, ML, time-series and data analysis for **quantitative research**.
- Processing HTML and PDF files and implementing **RAG** to extract structured data from the land use plan documents.
- Conducting **econometric** and **hedonic regression** analyses on property markets, housing development, and real estate dynamics.
- Acquiring GIS data from REST and ArcGIS Online endpoints in the Netherlands to construct a longitudinal panel dataset for research.
- Deploying and maintaining **Azure Functions** to automate web scraping of property listings and store real estate data for research purposes.
- Performing **GIS analysis** to produce and maintain geospatial research datasets and geocode real estate transactions from data sources such as MML, Ryhti, SYKE, and KVKL.
- Designing and deploying research databases using **PostgreSQL** and **DuckDB** to support team-wide data access.
- Building and managing **ETL/ELT pipelines** with data stewardship responsibilities to acquire, curate, and distribute research datasets.

Research Assistant Jun 2019 – Sep 2019
Aalto UniversityFull-time – Espoo, Finland Contributed to an **Agent-Based Modelling (ABM)** research project simulating urban mobility and parking pressure, developed in collaboration with **MIT CityLab**.

- Developed an **ABM simulation** to model parking pressure dynamics in the Otaniemi campus area.
- Collaborated with **MIT CityLab** on agent-based modelling of urban mobility patterns.
- Implemented simulations using the **GAMA Platform**.

Data Analytics Engineer Sep 2020 – Dec 2024
Qissa kaupunkisuunnitteluanalytiikka Oy – Startup *Part-time – Espoo, Finland*
Delivered data engineering and spatial analysis solutions for an urban analytics startup, supporting clients in urban planning and sustainability with end-to-end data workflows.

- Designed and maintained data pipelines to ingest **GIS data** from REST APIs, performing cleaning, **geocoding**, and storage in line with client specifications.
- Built a **spatial analytics dashboard** with a network analysis backend, integrating **HSL, HERE, and HSY APIs** to gather, process, and perform **spatial network analysis**.
- Extracted, processed, and served **census data** from the US, Denmark, Sweden, and Finland to calculate CO_2 emissions from commuting patterns.

Planning Assistant Nov 2018 – Jun 2021
WSP Finland Oy *Part-time – Helsinki, Finland*
Provided GIS and spatial data analysis support across urban planning and transport projects, delivering actionable insights and visualizations to multidisciplinary teams and external clients.

- Conducted **spatial data analysis**, processing and managing geospatial datasets across multiple concurrent projects.
- Performed **pedestrian network analysis** to forecast route choice and model post-development pedestrian flows, calibrated against real-world conditions.
- Prepared maps, graphs, and **data visualizations** for stakeholders, project managers, and clients to support planning decisions.
- Delivered **GIS analysis** to cross-disciplinary teams including transport engineers, landscape designers, urban planners, and architects.
- Configured **ArcGIS Online** platforms to support **maritime spatial planning** workflows.
- Developed dynamic **3D city models** using **ESRI CityEngine** to visualize urban development scenarios.

GIS Operator — Aug 2015 – May 2016
University of Tehran — *Part-time – Tehran, Iran*

- Collected, processed, and prepared **GIS datasets** to support urban designers in implementing and evaluating new urban design directives for Tehran.

Machine Learning | Statistical models | Econometrics

Uni-variate time-series:	●●●	**VAR time-series:**	●●●	**Co-integration time-series:**	●●
multi-variate time-series:	●●●	**OLS:**	●●●●●	**IV-OLS:**	●●●●
Panel data:	●●●●	**Generalised Method of Moments:**	●●●●	**Generalised linear models:**	●●
Spatial econometrics:	●●●●	**Survival/time-to-event:**	●●●●●	**Additive hazard models:**	●●●●●

Education

Aalto University — Espoo, Finland
Doctor of Philosophy in Technology, Real Estate Economics — *Sep 2020 - Present*

Aalto University — Espoo, Finland
MSc. (Tech.) Urban Studies and Planning in Real Estate Economic — *Sep 2018 - Jul 2020*
GPA: 4.1/5, Graduated with honours

University of Tehran — Tehran, Iran
BA. Town Planning — *Sep 2011 - Jun 2015*
GPA: 16.67/20, Academic excellence

Volunteer Work

- DataTribe collective moderator, contributor
- DataTribe collective speaker: Spatial data analysis, spatial data engineering
- Helsinki Data Week volunteer (2024 & 2025), contributor

Disclaimer: This resume is designed with LaTeX, and contains white text for the skill ratings to increase the accuracy of automated scanning and processing.